

Apiwit Karnjanavivin

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Education

Master of Science in Data Science, Artificial Intelligence, and Digital Business (M.Sc.) - First Class Honor June 2024 – September 2025
Gisma University of Applied Science
Potsdam, Germany
Final Dissertation: Better than it is: enhancing DeepSeekMoE architecture-based chatbots using dynamic routing algorithms

Bachelor of Engineering in Industrial Engineering (B.Eng.) August 2016 – April 2021
Kasetsart University
Bangkok, Thailand
Final Dissertation: Designing of lean-based hydroponic farm, its business plan, and financial analysis

Research Experience

Research Assistant - First author and main contributor (Volunteer) July 2025 – Present
Gisma University of Applied Science
Potsdam, Germany

- Proposed enhancing the DeepSeekMoE architecture by redesigning its routing algorithm, auxiliary load balance loss, and model training process to optimizing computational resource usage while retaining performance in training, fine-tuning, and inference.
- Co-authored a research paper targeted ACM publication, contributing introduction, foundations, related works, approach, and evaluation.
- Designed and executed ablation studies on the related routing algorithm, auxiliary load balance loss, and model architecture parameters on the prototype model with measurement metrics following research objectives to verify research claims and understand interactions between them.
- Contributing publication repository with ensure reproducibility for research paper publication of the baseline and prototype models with monitoring experiments, according to the objectives of the research.
- Collaborating on the research paper with a supervising professor through regular progress updates and iterative improvements based on professor’s feedbacks.

Professional Experience

In-Process Warehouse Leader October 2021 – November 2023
Raku Innovation Co., Ltd.
Bangkok, Thailand

- Directed Material Requirements Planning (MRP) and inventory audits across production warehouse and production lines using ERP systems (SAP B1).

Academic Projects

- A Survey of LLMs Architectures
- A Survey of LLMs Fine-Tuning Approaches
- A Survey of LLMs Fine-Tuning Optimization Approaches
- A Survey of Retrieval-Augmented Generation (RAGs)
- Hospital Stay Prediction - Machine Learning Operations (MLOps)
- Fine Tuning LLM - Large Language Model Operations (LLMOps)
- Retrieval Augmented Generation (RAG) - Large Language Model Operations (LLMOps)
- Adversarial Text Generation Against LLM Judges
- Memory Poison Audit for long-term LLM memory threats
- eBPF Gradient Aggregation for Distributed Machine Learning
- High-Throughput LLM Engine (Collective Communication Simulator)

Project documentation and code available at : <https://github.com/MisterFOURXXX?tab=repositories>

Achievements

- Social Media Extremism Detection Challenge - Top 24 percent of participants Kaggle Competition
- CIBMTR - Equity in post-HCT Survival Predictions Competition - Top 33 percent of participants
- PHEMS Hackathon: Pediatric Sepsis Prediction - Top 33 percent of participants
- Beyond Visible Spectrum: AI for Agriculture 2025 - Top 40 percent of participants
- FlightRank 2025: Aeroclub RecSys Cup - Top 45 percent of participants

Project documentation and code available at : <https://github.com/MisterFOURXXX/kaggle-competitions>

● Training and Certifications

Programming and Data Science Fundamentals

Coursera

- Python 3 Programming – University of Michigan
- Advanced Statistical Techniques for Data Science – Illinois Tech
- Git and GitHub Complete Master Class – Packt
- Applied Python Data Engineering – Duke University
- IBM Data Engineer – IBM

Machine Learning and Neural Network

- IBM Machine Learning – IBM
- IBM AI Engineering – IBM
- Foundations of Neural Networks – Johns Hopkins University
- Applied Machine Learning – Johns Hopkins University
- AI and Machine Learning Essentials with Python – University of Pennsylvania
- Machine Learning: Algorithms in the Real World – Alberta Machine Intelligence Institute

Deep Learning

- TensorFlow 2 for Deep Learning – Imperial College London
- IBM Deep Learning with PyTorch, Keras and Tensorflow – IBM
- Data Analytics and Deep Learning – Illinois Tech

Generative AI and Large Language Models (LLMs)

- Generative AI for Data Engineers – IBM
- Generative AI for Data Scientists – IBM
- Generative AI Engineering with LLMs – IBM
- IBM Generative AI Engineering – IBM
- IBM RAG and Agentic AI – IBM

Operations (MLOps and LLMOps)

- MLOps | Machine Learning Operations – Duke University
- Large Language Model Operations (LLMOps) – Duke University

● Skills and Proficiencies

Programming Languages

- Python, SQL, Shell, C, C++, Go, Git, LaTeX (Overleaf)

Data Science and Visualization

- Polars, Pandas, NumPy, Plotly, Seaborn, Matplotlib

Machine Learning, Deep Learning, and Large Language Models

- Scikit-learn, PyTorch, TensorFlow/ Keras, Hugging Face, Unsloth, LangChain, LangGraph, vLLM

Systems, Infrastructure, and Operations

- MLflow, DVC (Data Version Control), Oracle Cloud, AWS, Terraform, Docker, Kubernetes, FastAPI, Streamlit

Languages

- English (Fluent), Thai (Native), German (Beginner)

● References

Prof. Dr. Mohammad Mahdavi

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Gisma University of Applied Sciences, Professor

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Gisma University of Applied Sciences, Professor